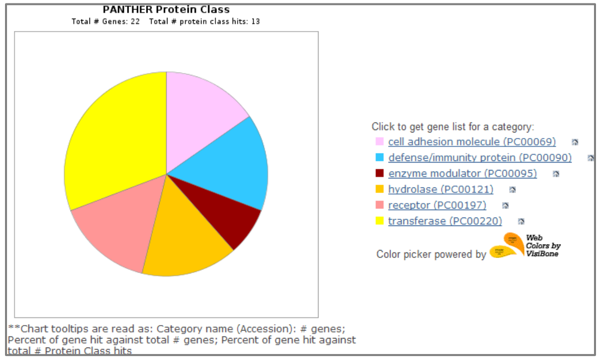
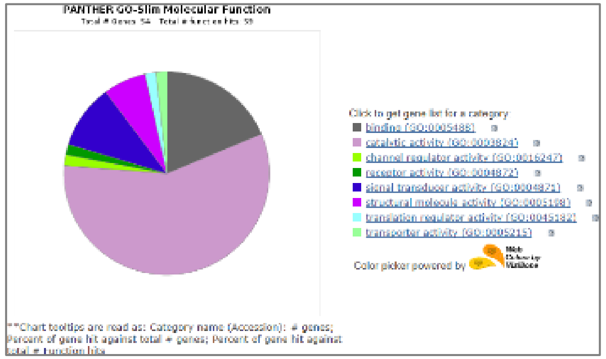
报告内容（功能注释 部分）

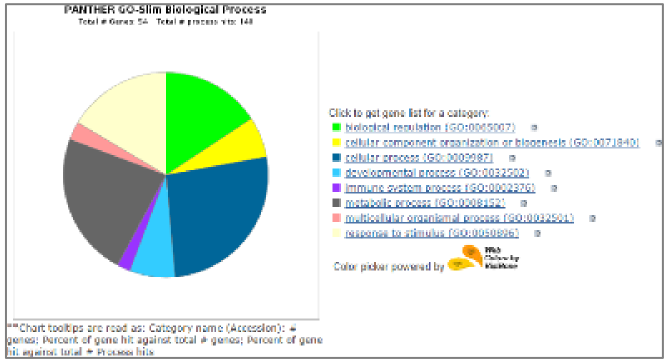
Entry	Protein names	Location	Function
P04626	Receptor tyrosine-p	**Isoform 1**: Cell membrane; Single-pass type I membrane p	Protein tyrosine kinase that **is part of several** cell surface receptor complexes, but that apparently **needs a coreceptor** for ligand bin
P00533	Epidermal growth fa	Golgi apparatus membrane; Single-pass type I membrane p	**Activating several signaling** cascades to convert extracellular cues. **Known ligands include** EGF, TGFA/TGF-alpha, amphiregulin
O14798	Tumor necrosis fact	Cell membrane; Lipid-anchor, GPI-anchor.	Receptor for the cytotoxic ligand TRAIL. Lacks a cytoplasmic death domain and hence **is not capable of inducing apoptosis**. May
Q01650	Large neutral amino	Cytoplasm, cytosol. Apical cell membrane; Multi-pass me	Sodium-independent, high-affinity transport of large neutral amino acids such as phenylalanine, tyrosine, leucine, arginine and tryptoph
P08637	Low affinity immun	Cell membrane {ECO:0000305}; Single-pass type I memb	**Receptor for the Fc region of IgG**. Binds complexed or aggregated IgG and also monomeric IgG. Mediates antibody-dependent cellula
P19440	Glutathione hydrola	Cell membrane {ECO:0000269|**PubMed:23682772**, ECO	Cleaves the gamma-glutamyl bond of extracellular glutathione (gamma-Glu-Cys-Gly), glutathione conjugates, and other gamma-glutamy
P35968	Vascular endothelial	Cell junction {ECO:0000250}. Endoplasmic reticulum {EC	Tyrosine-protein kinase that acts as a cell-surface receptor for VEGFA, VEGFC and VEGFD. **Plays an essential role in** the regulatio
P48551	Interferon alpha/bet	Isoform 1: Cell membrane {ECO:0000269|PubMed:76655	Associates with IFNAR1 to form the type I interferon receptor. Receptor for interferons alpha and beta. Involved in IFN-mediated ST
Q12884	Prolyl endopeptidas	Prolyl endopeptidase FAP: Cell surface {ECO:0000269|Pu	Involved in many cellular processes including tissue remodeling **(PubMed:17381073)**, fibrosis **(PubMed:17381073)**, wound healing, in
Q10471	Polypeptide N-acety	Golgi apparatus, Golgi stack membrane {ECO:0000269|Pu	**Catalyzes the initial reaction** in O-linked oligosaccharide biosynthesis, the transfer of an N-acetyl-D-galactosamine residue to a ser
O60603	Toll-like receptor 2	Membrane {ECO:0000250|UniProtKB:Q9QUN7}; Single-	**Cooperates with** LY96 to mediate the innate immune response to bacterial lipoproteins and other microbial cell wall components. Co



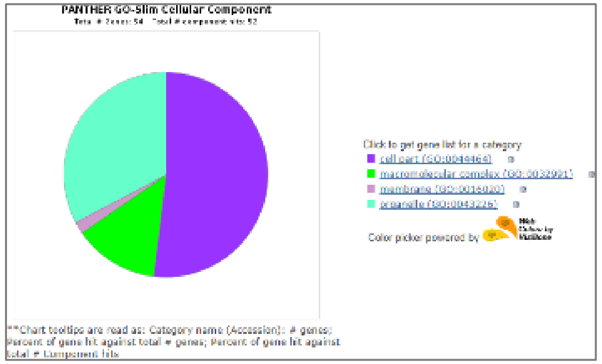
Protein Class



Molecular Function



Biological Process



Cellular Component

报告内容（原始数据 部分）

RawData1. SPR Raw Output

Time	Group_1	BG_1	Group_2	BG_2	Group_3	BG_3	Group_4	BG_4
551	480.97	138.70	477.37	140.04	470.82	133.16	469.54	76.34
552	539.99	105.76	547.88	107.97	552.77	102.02	545.38	77.34
553	541.54	29.19	548.64	22.41	545.01	18.57	545.43	78.34
554	536.69	41.69	536.40	43.58	536.96	51.25	532.00	79.34
555	533.39	43.22	529.40	46.63	524.02	48.55	523.06	80.34
556	478.57	54.51	481.72	60.77	476.71	57.07	479.30	81.34
557	473.95	22.18	469.19	24.16	463.57	27.52	464.75	82.34
558	487.82	10.28	492.97	5.64	499.54	11.98	502.71	83.34
559	536.03	40.17	534.00	36.22	533.41	28.86	539.87	84.34
560	549.67	21.87	552.65	16.51	559.56	9.84	561.14	85.34
561	558.26	19.73	554.06	15.15	556.97	11.38	561.11	86.34
562	552.98	28.16	547.82	34.09	543.64	30.48	539.31	87.34
563	534.92	24.01	542.53	27.09	542.07	31.30	534.46	88.34
564	570.37	24.61	572.91	18.99	566.50	13.11	561.07	89.34
565	569.48	78.92	568.55	82.52	560.86	76.02	556.15	90.34
566	555.84	69.76	555.21	75.11	547.83	81.54	543.95	91.34
567	559.58	48.72	565.60	41.69	566.88	34.24	563.05	92.34
568	528.31	13.33	520.94	5.36	525.90	11.10	520.36	93.34
569	522.80	36.81	521.79	44.34	527.12	41.26	531.39	94.34
570	509.82	60.00	503.06	59.09	498.80	55.40	500.03	95.34
571	521.27	32.24	529.07	36.81	522.84	37.89	529.81	96.34
572	525.11	14.13	524.26	13.02	531.77	20.46	532.72	97.34
573	524.00	1.31	529.53	-2.88	526.67	4.55	532.34	98.34
574	524.77	101.78	529.46	100.20	531.08	104.67	537.96	99.34
575	510.68	106.67	508.07	112.56	504.74	118.11	510.26	100.34
576	534.90	113.99	533.89	115.23	532.26	123.02	524.92	101.34
577	527.20	44.44	528.46	44.64	531.60	37.58	533.35	102.34
578	519.79	21.26	518.34	27.61	518.59	34.77	515.72	103.34
579	507.37	12.30	501.30	7.59	503.82	13.34	504.09	104.34
580	527.41	32.85	534.04	29.61	541.81	35.64	546.93	105.34

Raw Value Record

RawData2. MS Raw Output

返回上一级 | 全部文件 > 我的资源 > RAWDATA

文件名 ↓

- MSRAW20180304_FingerPrint_AS7H63KKSIG8239WEJG
- MSRAW20180210_FingerPrint_DOFU74HSIIQ8AJNVAA112
- MSRAW20180206_FingerPrint_YFG786A8723HJFQAUI
- MSRAW20180205_FingerPrint_V825ECS028RHGW1AVN1
- MSRAW20180205_FingerPrint_AH8YJ4520FMAORUSHKF03
- MSRAW20180121_FingerPrint_GH8R36X5BDW1Q97KOI
- MSRAW20180111_FingerPrint_GH8R36X5BDW1Q97KOI
- MSRAW20180104_FingerPrint_AD84GWH757XFA510

#示例：https://**eyun**.baidu.com/s/3ggUkt59 密码：******

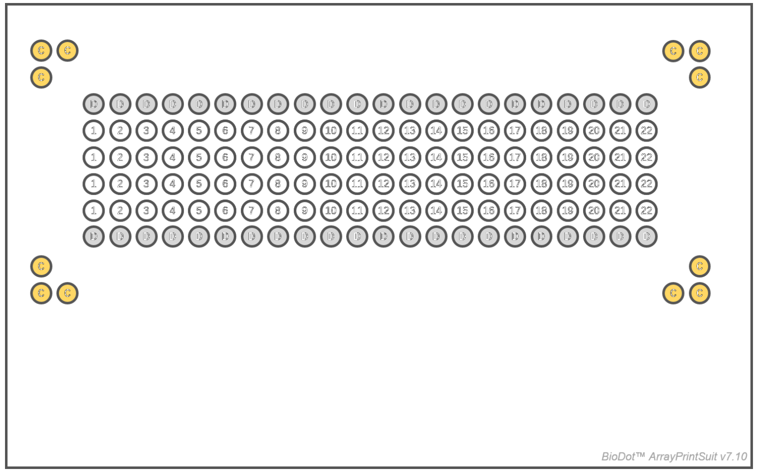
RawData3. Function Annotation

	Gene ID	IDs	Pathway	MF	BP	CC
1	HUMAN|HGNC=3619|UniProtKB=P08637	P08637	IgG binding(GO:0019864)	immune response(GO:0006955);Fc-gamma receptor signaling pathway involved	plasma membrane(GO:0005886);external side of plasma membrane(GO:000	Immunoregulatory interactions between a Lymphoid and a non-Lymphoid cell;Immune System;Regulation of
2	HUMAN|HGNC=4124|UniProtKB=Q10471	Q10471	polypeptide N-acetylgalactosaminyltransferase activity(GO:0004653);protein	immunoglobulin biosynthetic process(GO:0002378);protein O-linked glycosylation(GO:00	Golgi membrane(GO:0000139);endoplasmic reticulum membrane(GO:000	Golgi-to-ER retrograde transport;Post-translational protein modification;Metabolism of proteins;Membrane
3	HUMAN|HGNC=3689|UniProtKB=P21802	P21802	protein kinase activity(GO:0004672);protein tyrosine kinase activity(GO:000471	negative regulation of transcription by RNA polymerase II(GO:0000122);MA	nucleus(GO:0005634);nucleoplasm(GO:0005654);cytoplasm(GO:0005737);Golgi	Signaling by FGFR2 IIIa TM;Signaling by FGFR2 fusions;Signaling by FGFR2 in disease;Disease;Signaling
4	HUMAN|HGNC=4127|UniProtKB=Q7Z7M9	Q7Z7M9	polypeptide N-acetylgalactosaminyltransferase activity(GO:0004653);carbohydrate	glycosaminoglycan biosynthetic process(GO:0006024);O-glycan processing(GO:001	Golgi membrane(GO:0000139);cellular_component(GO:0005575);integral	Post-translational protein modification;O-linked glycosylation;Metabolism of proteins;O-linked glycosylation of mucins
5	HUMAN|HGNC=11848|UniProtKB=O60603	O60603	lipopolysaccharide binding(GO:0001530);lipopolysaccharide receptor activity(GO:000187	response to hypoxia(GO:0001666);microglial cell activation(GO:0001774);cell	cytoplasm(GO:0005737);Golgi apparatus(GO:0005794);plasma membrane(GO:000	Class I MHC mediated antigen processing & presentation;Toll Like Receptor 4 (TLR4) Cascade;Disease;Defensin
6	HUMAN|HGNC=11772|UniProtKB=P36897	P36897	protein kinase activity(GO:0004672);protein serine/threonine kinase	activation of MAPKK activity(GO:0000186);skeletal system development(GO:0	intracellular(GO:0005622);cell(GO:0005623);endosome(GO:0005768);plasma membrane(GO:000	SMAD2/3 Phosphorylation Motif Mutants in Cancer;Disease;Post-translational protein
7	HUMAN|HGNC=5433|UniProtKB=P48551	P48551	type I interferon receptor activity(GO:0004905);protein binding(GO:000551	cell surface receptor signaling pathway(GO:0007166);JAK-STAT cascade(GO:00072	extracellular region(GO:0005576);extracellular space(GO:0005615);intracellular(GO:0	
8	HUMAN|HGNC=3765|UniProtKB=P36888	P36888	transmembrane receptor protein tyrosine kinase activity(GO:0004714);cytokine	leukocyte homeostasis(GO:0001776);myeloid progenitor cell differentiation(GO:	nucleus(GO:0005634);endoplasmic reticulum(GO:0005783);endoplasmic reticulum	Immune System;Signaling by Interleukins;Cytokine Signaling in Immune system
9	HUMAN|HGNC=8804|UniProtKB=P09619	P09619	protein tyrosine kinase activity(GO:0004713);platelet activating factor	MAPK cascade(GO:0000165);glycosaminoglycan biosynthetic process(GO:000602	nucleus(GO:0005634);cytoplasm(GO:0005737);plasma membrane(GO:0005886);focal	FCERI mediated MAPK activation;FRS-mediated FGFR2 signaling;Signaling by SCF-KIT;PI5P, PP2A and IER3 Regulate
10	HUMAN|HGNC=11906|UniProtKB=O14798	O14798	transmembrane signaling receptor activity(GO:0004888);tumor necrosis factor-activated	inflammatory response(GO:0006954);immune response(GO:0006955);signal	plasma membrane(GO:0005886);integral component of plasma	TP53 Regulates Transcription of Cell Death Genes;TP53 Regulates Transcription of Death Receptors and

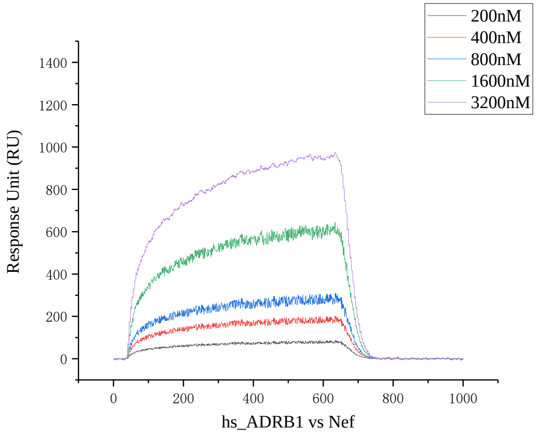
Database

质控手段

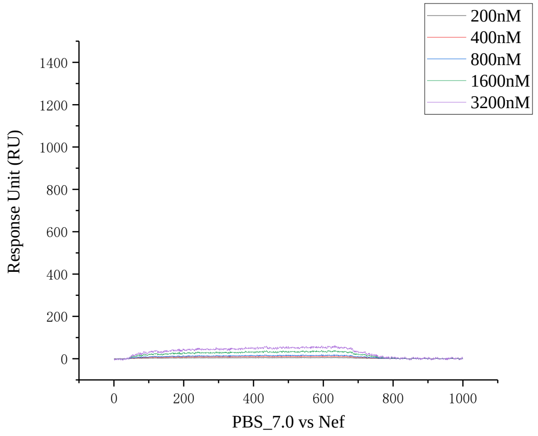
QA1：点样——设备可视监控

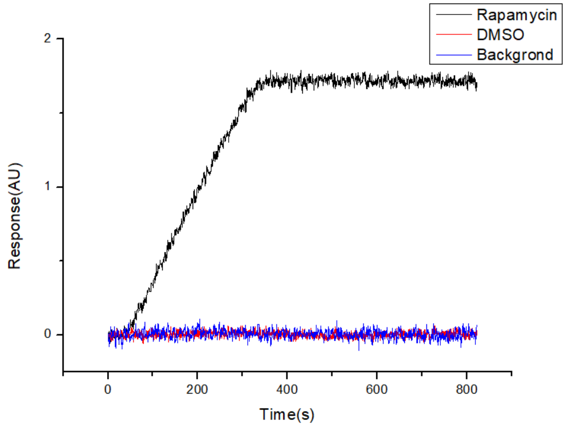


QA2：浓度梯度（平行）

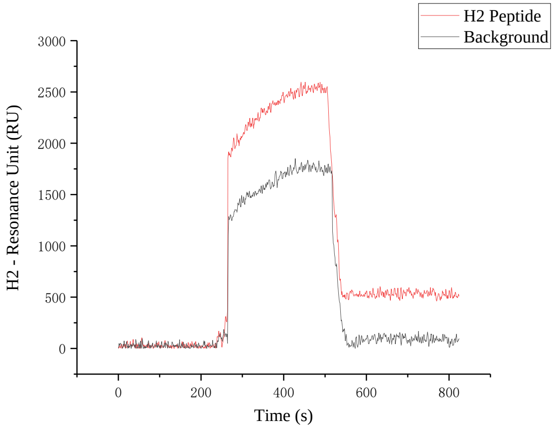


QA3：阴性对照





QA4：系统对照

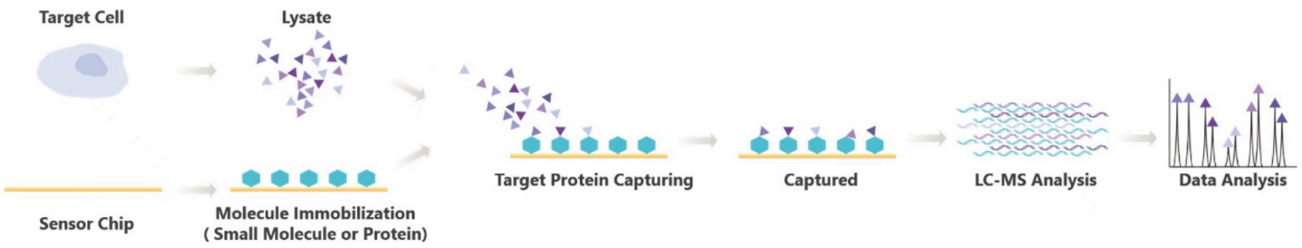


QA5：载量及质谱

（Filter: >550 RU per kilo-pMSMs）

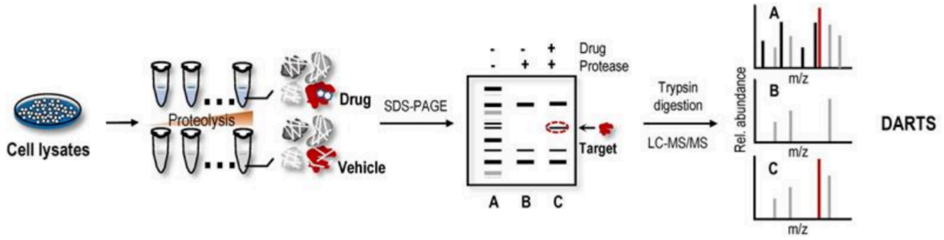
靶标捕获技术 与前代技术的对比

第二代：基于SPR的靶标蛋白组学（靶标捕获，SPR-LCMS，2015年面市）：



SPR-LCMS流程概括：
“芯片打印→捕获→ 洗涤分离→酶解→质谱”

第一代：药物亲和反应靶向稳定性技术服务（DARTS，2009年面市）：



DARTS流程概括：
“直接酶解→捕获→跑胶分离→质谱”

★ 性能对比：

	引入标记	反应载体	除杂方式	靶向的实质	结果范围	避免竞争结合	费用
SPR-LCMS	无	固相芯片	亲和预洗涤 (自动)	完整蛋白	仅蛋白	实时监控	2.4~2.8w
DARTS	无	无	跑胶 (肉眼) 或无	酶解后肽段	酶解后肽段	否	3.0~4.2w

案例1: 老药新用的药效机制解析

Clinical Cancer Research

AACR American Association for Cancer Research

Discovery of berberine that targetedly induce autophagic degradation of both BCR-ABL and BCR-ABL T315I through recruiting LRSAM1 for overcoming imatinib-resistance

Zhao Yin, Guiping Huang, Chunming Gu, et al.

Clin Cancer Res Published OnlineFirst February 25, 2020.

Abstract

Purpose: Imatinib, the breakpoint cluster region protein (BCR)/Abelson murine leukemia viral oncogene homolog (ABL) inhibitor, is widely used to treat chronic myeloid leukemia (CML). However, imatinib resistance develops in many patients. Therefore, new drugs with improved therapeutic effects are urgently needed. Berberine (BBR) is a potent BCR-ABL inhibitor for imatinib-sensitive and -resistant CML.

Experimental design: Protein structure analysis and virtual screening were used to identify BBR targets in CML. Molecular docking analysis, surface plasmon resonance imaging (SPRi), nuclear magnetic resonance (NMR) assays, and thermoshift assays were performed to confirm the BBR target. The change in BCR-ABL protein expression after BBR treatment was assessed by western blotting. The effects of BBR were assessed in vitro in cell lines, in vivo in mice, and in human CML bone marrow cells as a potential strategy to overcome imatinib resistance.

Results: We discovered that BBR bound to the protein tyrosine kinase (PTK) domain of BCR-ABL. BBR inhibited the activity of BCR-ABL and BCR-ABL with the T315I mutation, and it also degraded these proteins via the autophagic lysosome pathway by recruiting E3 ubiquitin-protein ligase LRSAM1. BBR inhibited the cell viability and colony formation of CML cells and prolonged survival in CML mouse models with imatinib sensitivity and resistance.

Conclusions: The results show that BBR directly binds to and degrades BCR-ABL and BCR-ABL T315I via the autophagic lysosome pathway by recruiting LRSAM1. The use of BBR is a new strategy to improve the treatment of CML patients with imatinib sensitivity or resistance.

靶标捕获

↓

验证结合

↓

寻找结合表位

↓

验证结合表位

↓

治疗作用及下游机制

药物靶点

构效关系

案例1: 老药新用的药效机制解析

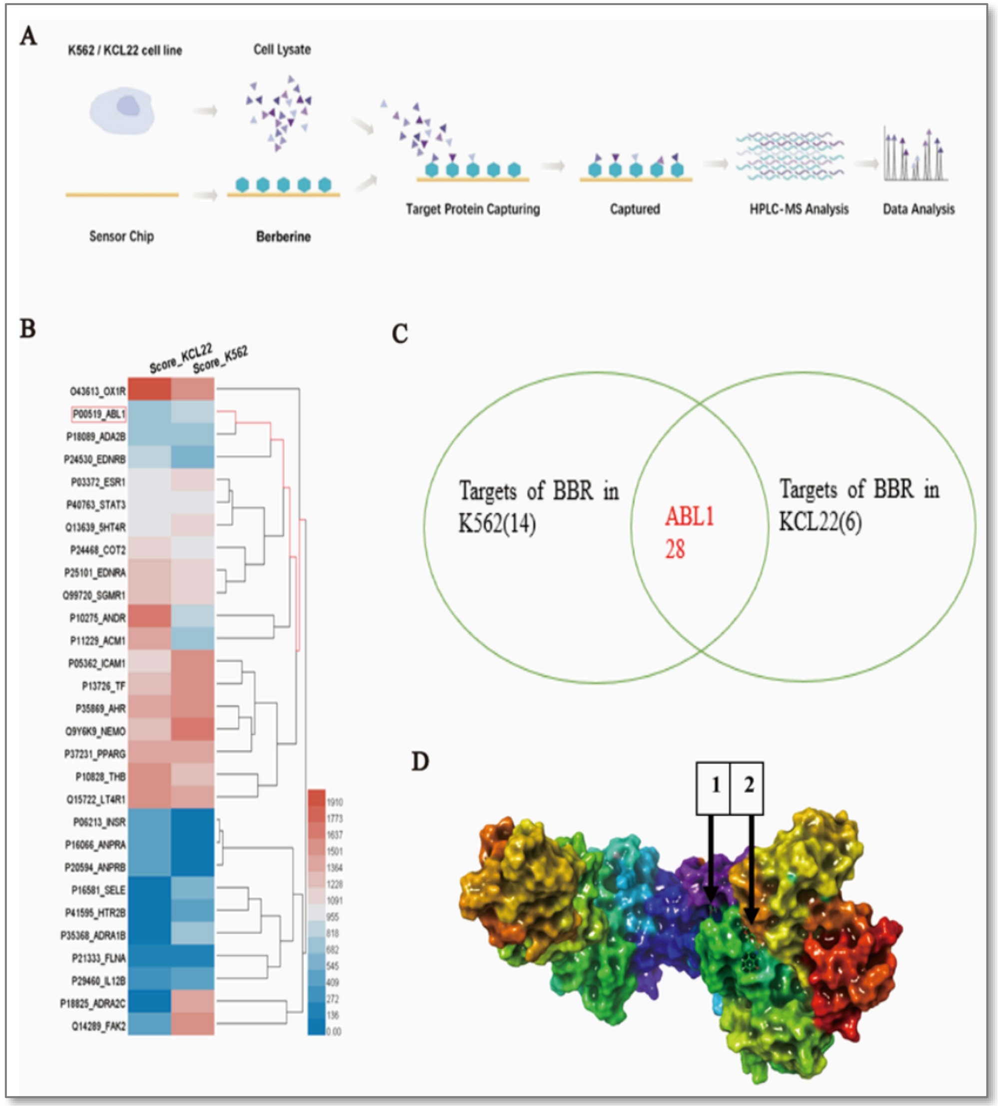
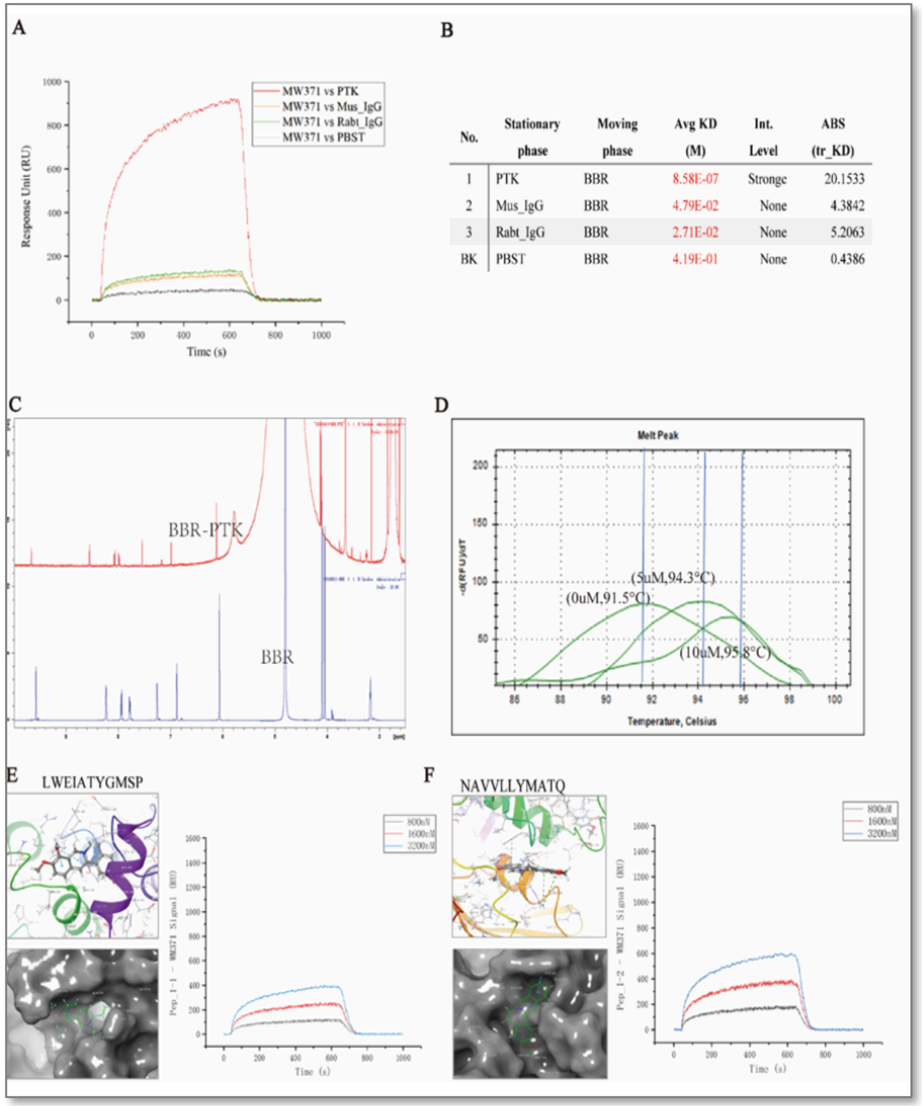


Figure.1



No.	Stationary phase	Moving phase	Avg KD (M)	Int. Level	ABS (tr_KD)
1	PTK	BBR	8.58E-07	Stronge	20.1533
2	Mus_IgG	BBR	4.79E-02	None	4.3842
3	Rabt_IgG	BBR	2.71E-02	None	5.2063
BK	PBST	BBR	4.19E-01	None	0.4386

Figure.2

靶标捕获

验证结合

寻找结合表位

验证结合表位

治疗作用及

下游机制

药物靶点

构效关系

案例2: 药效的多靶点效应解析

RESEARCH ARTICLE Open Access

Identification of berberine as a novel drug for the treatment of multiple myeloma via targeting UHRF1

Chunming Gu[1,2†], Zhao Yin[1,2†], Hong Nie[3,4†], Yanjun Liu[1], Juhua Yang[1], Guiping Huang[1], Jianping Shen[5*], Liguo Chen[2*] and Jia Fei[1,2*]

Abstract

Background: Current therapies for multiple myeloma (MM) are associated with toxicity and resistance, highlighting the need for novel effective therapeutics. Berberine (BBR), a botanical alkaloid derived from several Berberis medicinal plants, has exhibited anti-tumor effects, including against multiple myeloma (MM); however, the molecular mechanism underlying the anti-MM effect has not been previously described. This study aimed to identify the target of berberine and related mechanisms involved in its therapeutic activity against MM.

Results: Here, we demonstrated that BBR treatment killed MM cells in vitro and prolonged the survival of mice bearing MM xenografts in vivo. A screening approach integrating surface plasmon resonance (SPR) with liquid chromatography-tandem mass spectrometry (LC-MS/MS) identified UHRF1 (ubiquitin-like with PHD and RING Finger domains 1) as a potential target of BBR. Combining molecular docking and SPR analysis, we confirmed UHRF1 as a BBR-binding protein and discovered that BBR binds UHRF1 in the tandem tudor domain and plant homeodomain (TTD-PHD domain). BBR treatment induced UHRF1 degradation via the ubiquitin-dependent proteasome system and reactivated p16^{INK4A} and p73 in MM cells. Overexpression of UHRF1 promoted the MM cell proliferation and rendered MM cells more resistant to BBR, while silencing of UHRF1 with siRNA attenuated BBR-induced cytotoxicity.

Conclusions: In summary, our study has identified UHRF1 as a direct target of BBR and uncovered molecular mechanisms involved in the anti-MM activity of BBR. Targeting UHRF1 through BBR may be a novel therapeutic strategy against MM.

Keywords: Berberine, Multiple myeloma, SPR-LC-MS/MS approach, UHRF1, Target identification

靶标捕获

↓

验证结合

↓

寻找结合表位

↓

验证结合表位

↓

治疗作用及

下游机制

药物靶点

构效关系

案例2: 药效的多靶点效应解析

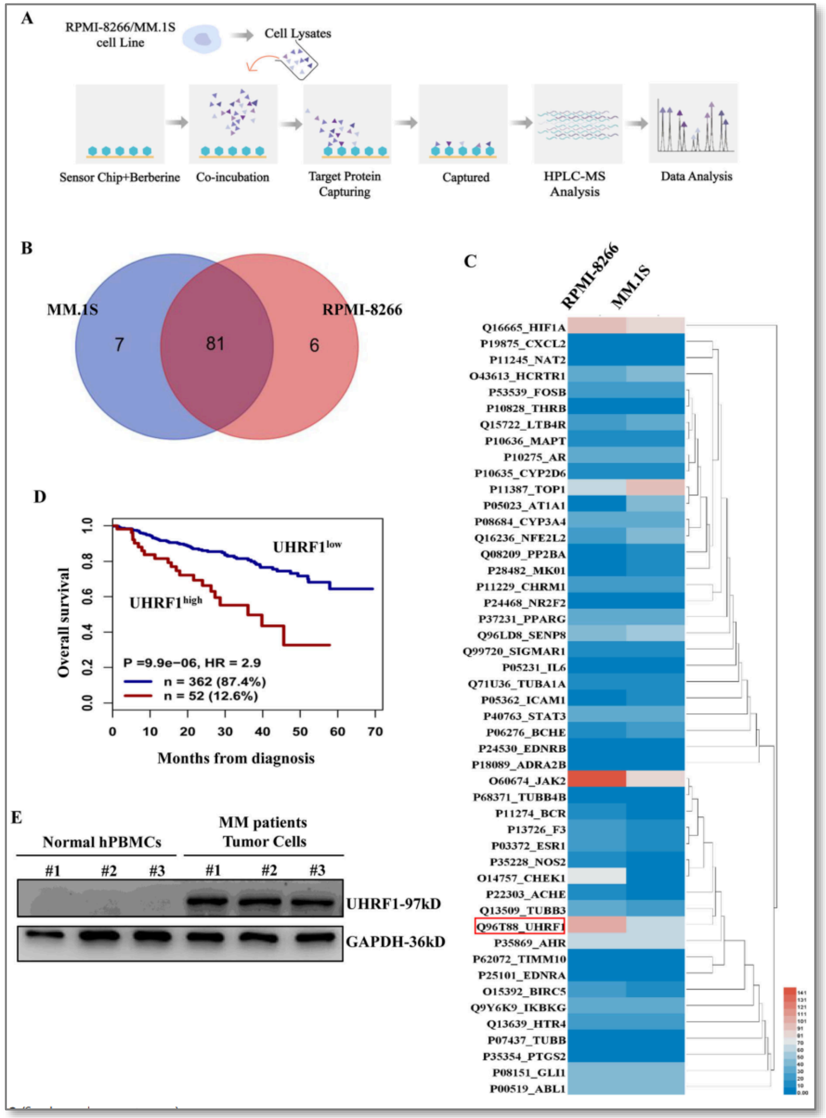


Figure.1

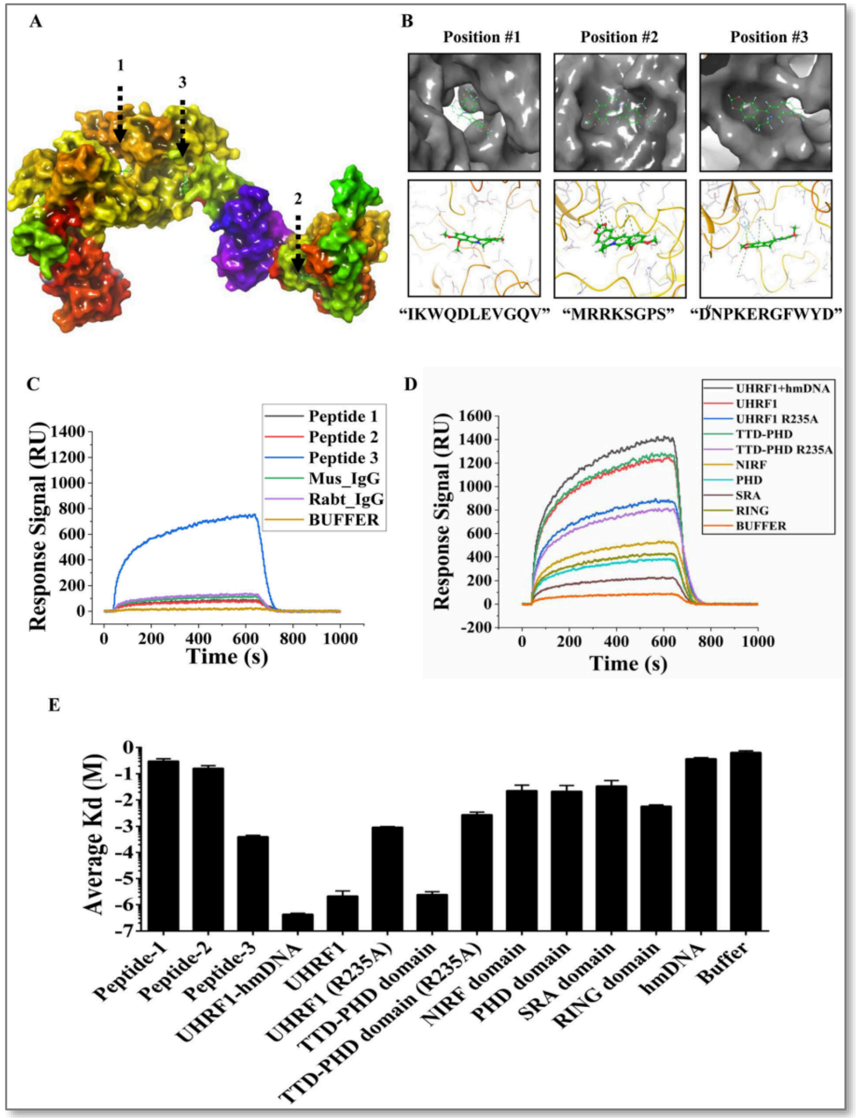


Figure.2

靶标捕获

验证结合

寻找结合表位

验证结合表位

治疗作用及

下游机制

药物靶点

构效关系

案例3: 原研药物的药效解析

中国医学科学院 北京协和医学院 医药生物技术研究所
INSTITUTE OF MEDICINAL BIOTECHNOLOGY, CHINESE ACADEMY OF MEDICAL SCIENCES & PEKING UNION MEDICAL COLLEGE

nature COMMUNICATIONS

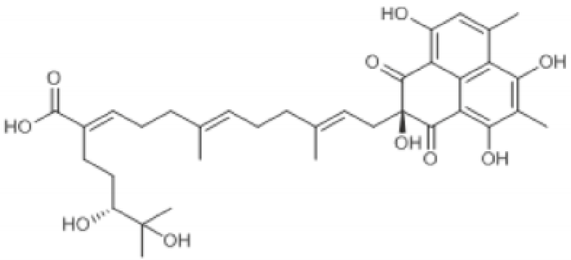
ARTICLE

Check for updates

https://doi.org/10.1038/s41467-022-29690-x OPEN

An anti-influenza A virus microbial metabolite acts by degrading viral endonuclease PA

Jianyuan Zhao[1,5], Jing Wang[1,5], Xu Pang[1,5], Zhenlong Liu[2], Quanjie Li[1], Dongrong Yi[1], Yongxin Zhang[1], Xiaomei Fang[1], Tao Zhang[1], Rui Zhou[1], Tao Zhang[1], Zhe Guo[1], Wancang Liu[1], Xiaoyu Li[1], Chen Liang[2], Tao Deng[3], Fei Guo[4]✉, Liyan Yu[1]✉ & Shan Cen[1]✉



APL-16-2

Canddiates of APL-16-5 Molecular Capture	
Abbreviation	Score
KEAP1	1221.95
HERC5	1023.75
TRIM25	712.22
ISG15	703.29
UBE2E2	654.96

案例4: 天然提取物的药效机制解析

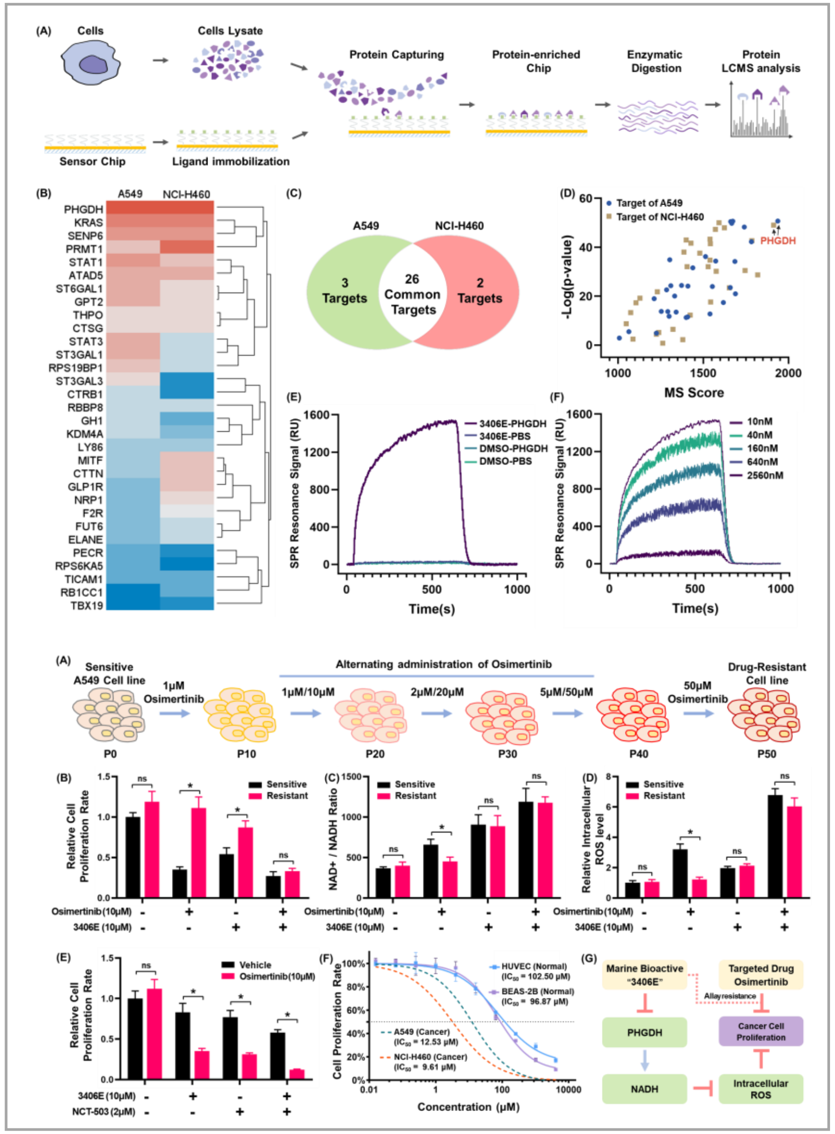


Figure.1

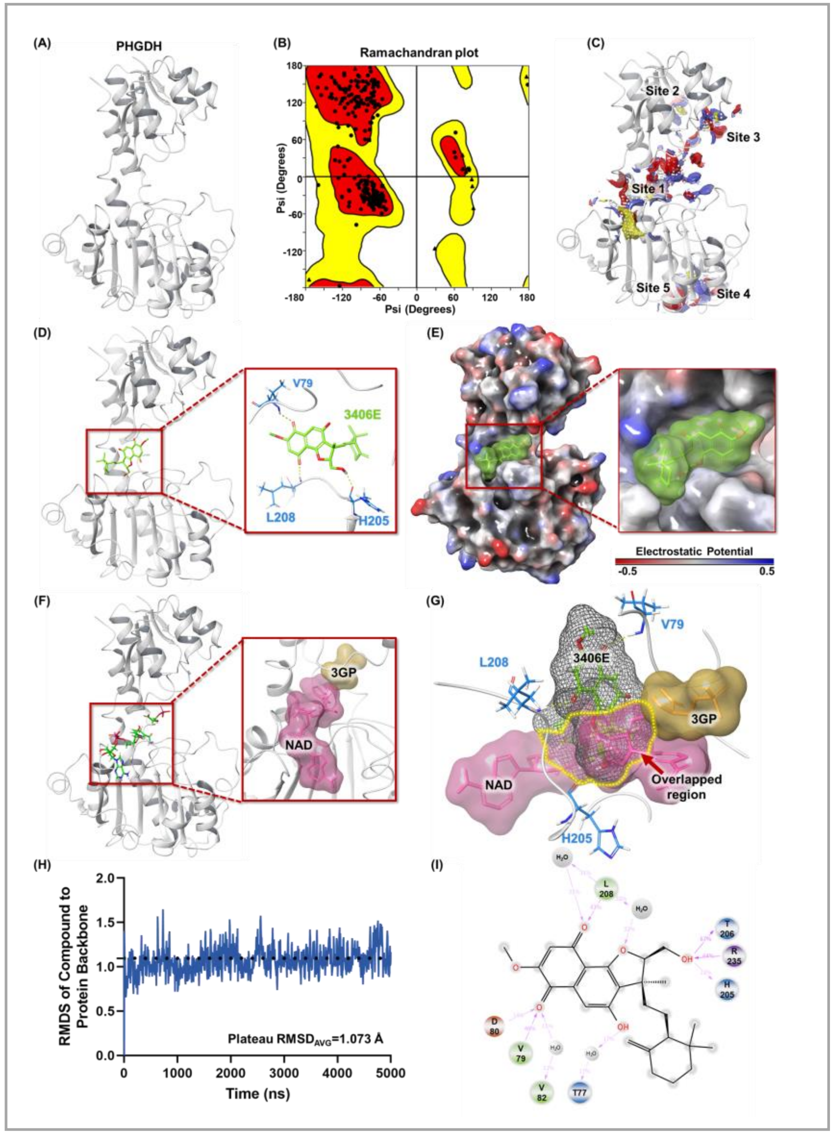


Figure.2

中科院南海海洋所
鞠建华 研究员 团队

海洋天然产物提取

药效靶标分析

结合动力学分析

药效机制验证

治疗作用

研究流程总结

研究目的	研究手段	实现方式
Step 1: 确认药物靶点	靶标捕获	平台1：靶标捕获
	↓ 验证结合	平台2：互作分析
Step 2: 亚结构药效动力学	↓ 寻找结合表位	平台6：分子动力学
	↓ 验证结合表位	平台7：互作分析
	↓ 阐明药物分子机制	

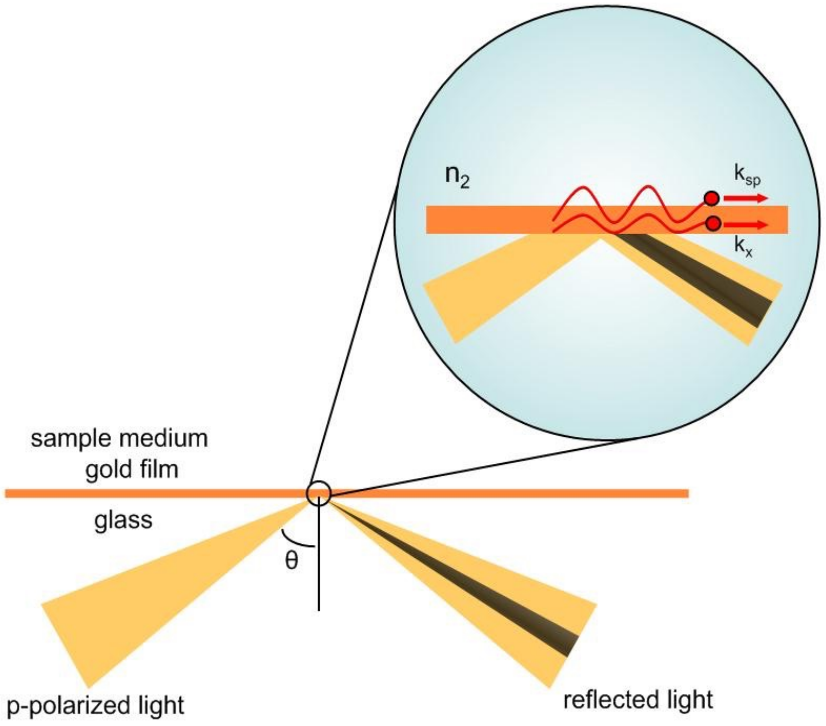
研究方向

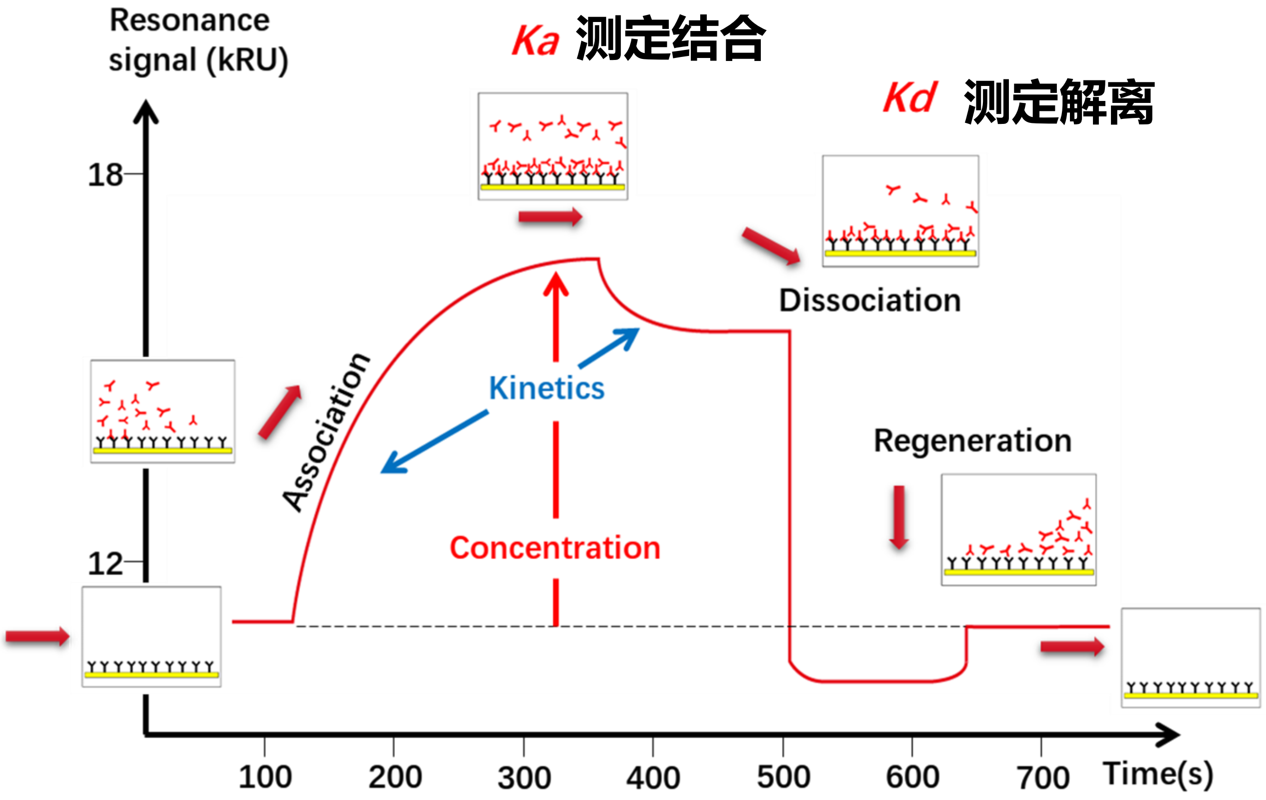
- 已知药物，分析靶蛋白（ 药物 → 靶蛋白 ）
  - 平台1：靶标捕获
  - 平台2：互作分析
- 已知靶蛋白，分析药物（ 靶蛋白 → 药物 ）
  - 平台3：高通量对接筛选
  - 平台4：高通量实体库筛选
- 已知药物及靶蛋白，分析亚结构药效机制（ 靶 药 → 机制 ）
  - 平台5：表位捕获平台
  - 平台6：药效分子动力学运算平台
  - 平台7：药物口袋反向验证平台

Surface Plasmon Resonance image, SPRi

（表面等离子共振成像，第二代SPR技术）

SPRi 技术示意图





亲和力分析作为互作金标准

组学分析		验证手段
靶标捕获 (靶标蛋白组学)	→	亲和力分析
转录组学测序	→	RT-qPCR
定量蛋白组学	→	Western blot

多种 亲和力验证手段 的 优势对比

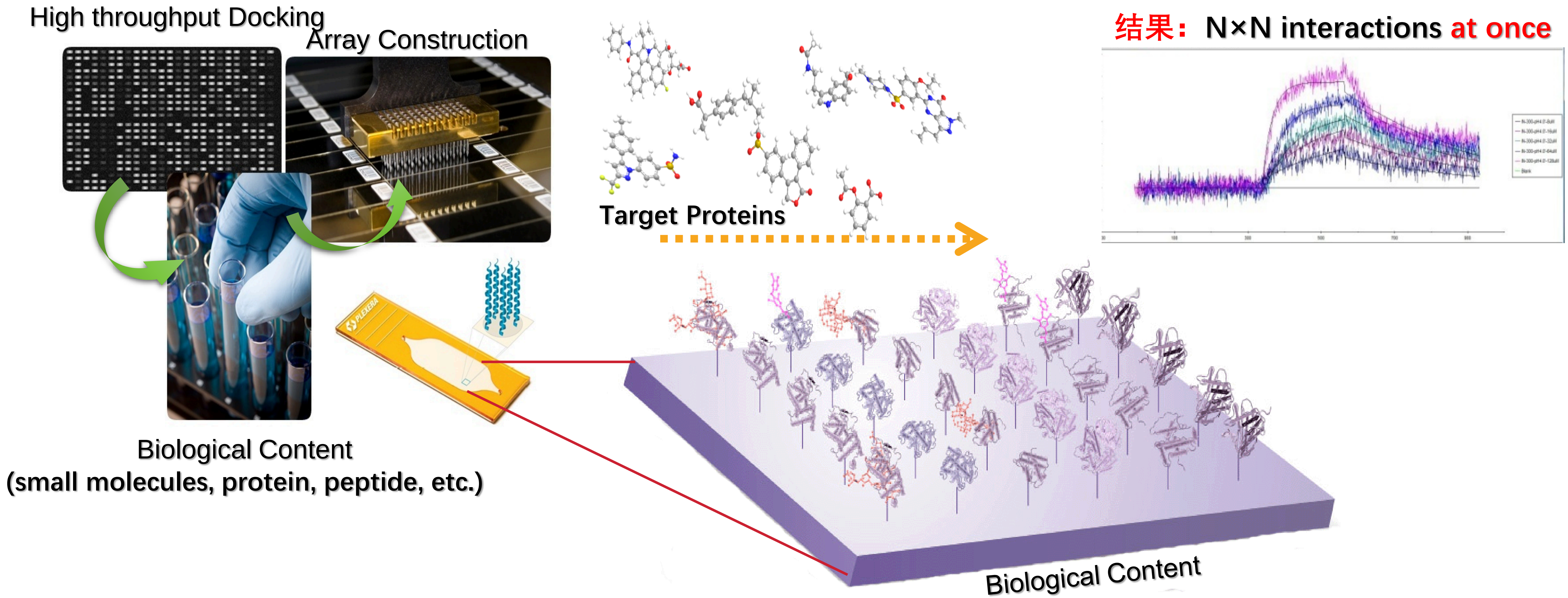
	SPR	ITC	MST	BLI
COMPARISON OF BIOMOLECULAR INTERACTION ANALYSIS METHODS				
	SPR-Surface Plasmon Resonance Reichert Systems	ITC-Isothermal Titration Calorimetry	MST-Microscale Thermophoresis	BLI-Biolayer Interferometry
Signal read-out (change of...)	Refractive index	Temperature	Fluorescence intensity	Wavelength shift
Kinetics	Yes	No	No	Protein-Protein Only
Affinity	Yes	Yes	Yes	Yes
Thermodynamics	Yes	Yes	Yes	Limited
Stoichiometry	Yes	Yes	Yes	Yes
Concentration – Analyte	Yes	No	No	Yes
Affinity range	pM - mM	nM - µM	pM – mM	nM - µM
Precision of read-out	High	Medium	Low	Medium
Sensitivity	High	Low	Medium	Low
Temperature Range	10°C below ambient – 70°C	2-80°C	20-45°C	Ambient + 4°C
Sample Consumption	Low	High	Very Low	Low
Sample Types *	Maximum Flexibility	Few	Intermediate	Intermediate
Buffers	No restrictions for aqueous solutions Concentrated Acids/Bases Most organic solvents	Aqueous solutions with small amounts of organic solvents	No restrictions for aqueous solutions (incl. 8M urea, 4M MgCl2 etc.)	Aqueous solutions with small amounts of organic solvents
Maintenance	Operator or Service Tech	Service technician	NA	Service technician
Advantages to Methods	Highest information content Reference method for FDA/ICH/EMA Max experimental flexibility	No immobilization No labeling	Low sample consumption	Highest throughput
Disadvantages to Methods	Immobilization of one binding partner required. Trained personnel is required for high quality data	High sample Consumption Limited applicability - reactions with measureable temperature change.	Cannot discern 2nd binding site or non-specific binding Labeling with hydrophobic fluorophores required, which can alter binding profile. High CV of fluorescence data.	Immobilization of one binding partner required. Dissociation phases are imprecise due to analyte rebinding (no flow through system)

CHART COLOR KEY: Favorable performance | Moderate performance | Unfavorable performance

* See table below for sample types

数据来源：**Thermo Fisher, 2019**

高通量相互作用筛选


High throughput Docking
Array Construction
Biological Content
(small molecules, protein, peptide, etc.)
Target Proteins
结果：N×N interactions at once
Biological Content

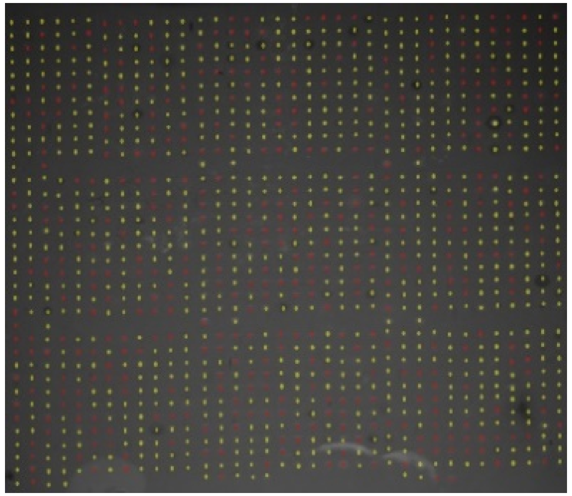
案例4: 亲和力筛选

916种药物小分子与11个药物靶标蛋白的相互作用

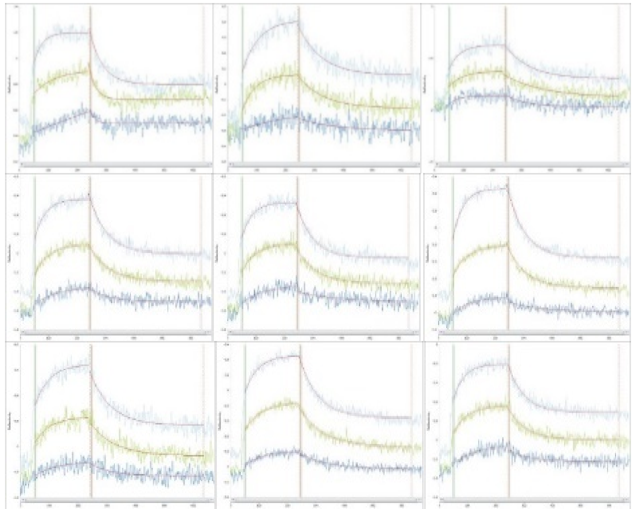
样本：916药物衍生物（仿制药前体）

分子库：11蛋白药效相关蛋白

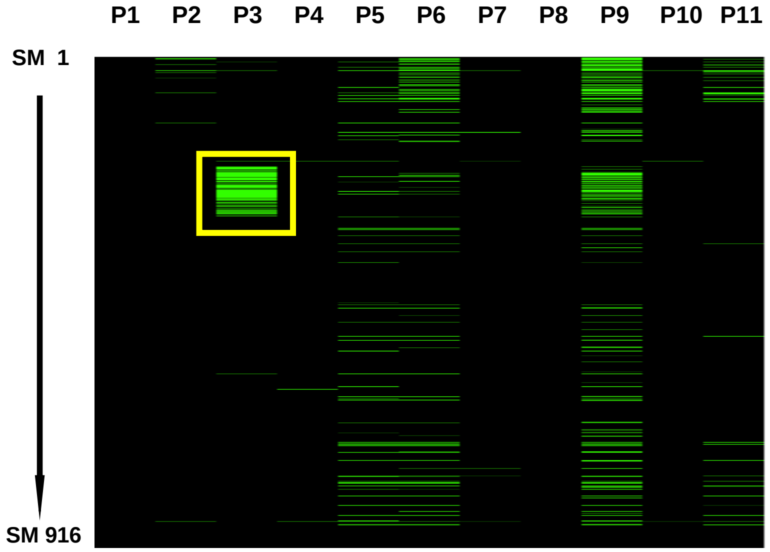
合作者：日本武田制药株式会社



芯片表面动态扫描



亲和力数据曲线



热图（互作强度）

案例5: 对比不同抑制剂与靶蛋白的结合强弱关系

No.	固定相（打印于芯片表面）	流动相（流经芯片表面）	Avg ka (1/Ms)	Avg kd (1/s)	Avg KD (M)	Int.Intensity Level	ABS (tr_KD)
1-1	Acarbose	α-Glucosidase	1.43E+02	1.04E-05	7.22E-08	**Strong**	23.723
1-2	Apigenin	α-Glucosidase	1.47E+05	3.80E-03	2.58E-08	**Strong**	25.208
1-3	Clitorin	α-Glucosidase	1.71E+03	1.00E-01	5.85E-05	**middle**	14.060
1-4	Genkwanin	α-Glucosidase	1.03E+01	6.58E-02	6.42E-03	**Weak**	7.283
1-5	Hyperoside	α-Glucosidase	9.83E+00	1.22E-05	1.24E-06	**Strong**	19.625
1-6	Isorhamnetin	α-Glucosidase	2.16E+02	3.88E-05	1.80E-07	**Strong**	22.405
1-7	Kaempferide	α-Glucosidase	2.54E+01	3.01E-01	1.18E-02	**Weak**	6.400
1-8	Kaempferol	α-Glucosidase	1.45E+02	4.00E-03	2.76E-05	**middle**	15.144
1-9	Luteolin	α-Glucosidase	9.77E+02	1.34E-03	1.37E-06	**Strong**	19.474
1-10	Isoquercitrin	α-Glucosidase	5.30E+00	2.66E-02	5.03E-03	**Weak**	7.636
1-11	Myricetin	α-Glucosidase	3.81E+04	8.19E-03	2.15E-07	**Strong**	22.150
1-12	Myricetrin	α-Glucosidase	9.65E+00	1.22E-01	1.27E-02	**Weak**	6.303
1-13	S-Naringenin	α-Glucosidase	5.59E+02	1.34E-03	2.40E-06	**Strong**	18.666
Control	DMSO	α-Glucosidase	1.16E+00	6.82E-01	5.86E-01	**VW/None**	0.770

案例6: 药物-蛋白亲和力测定及筛选

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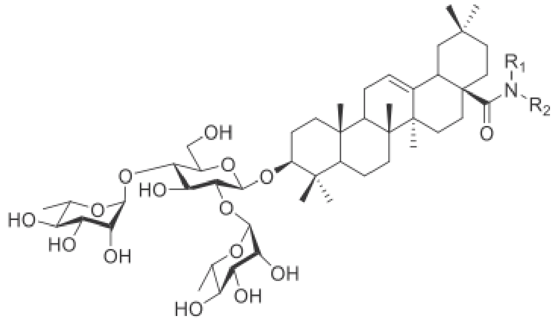
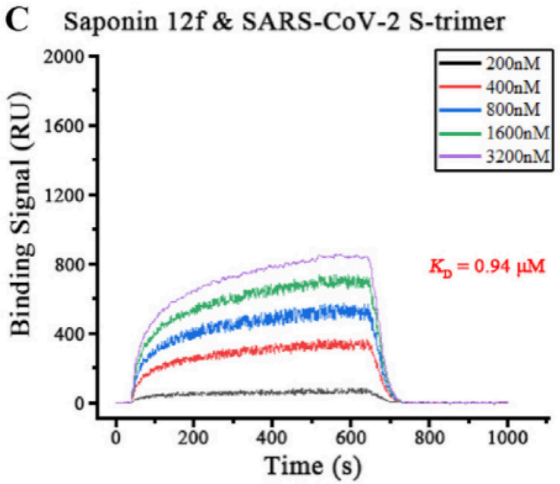


Discovery and structural optimization of 3-*O*-β-chacotriosyl oleanane-type triterpenoids as potent entry inhibitors of SARS-CoV-2 virus infections

Hui Li [a, 1], Chen Cheng [b, 1], Sumei Li [c, 1], Yan Wu [d], Zhihao Liu [a], Mingjian Liu [a], Jianxin Chen [e], Qiuyu Zhong [a], Xuesha Zhang [a], Shuwen Liu [b, f, **], Gaopeng Song [a, *]

[a] Key Laboratory for Biobased Materials and Energy of Ministry of Education, College of Materials and Energy, South China Agricultural University, Guangzhou, 510642, China
[b] Guangdong Provincial Key Laboratory of New Drug Screening, School of Pharmaceutical Sciences, Southern Medical University, Guangzhou, 510515, China
[c] Department of Human Anatomy, School of Medicine, Jinan University, Guangzhou, 510632, China
[d] State Key Laboratory of Virology, Wuhan Institute of Virology, Center for Biosafet Mega-Science, Chinese Academy of Sciences, Wuhan, 430071, China
[e] Guangdong Provincial Key Laboratory of Veterinary Pharmaceutics Development and Safety Evaluation, College of Veterinary Medicine, South China Agricultural University, Guangzhou, 510642, China
[f] State Key Laboratory of Organ Failure Research, Guangdong Provincial Institute of Nephrology, Southern Medical University, Guangzhou, 510515, China

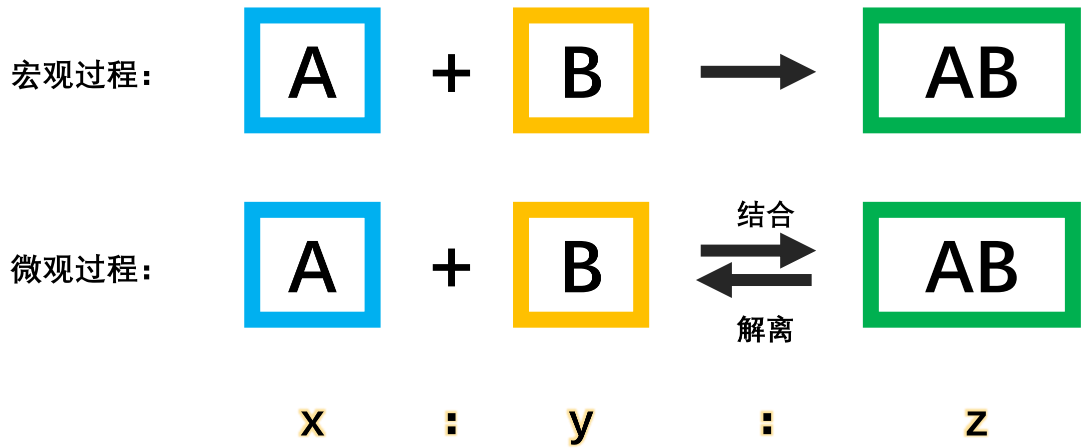


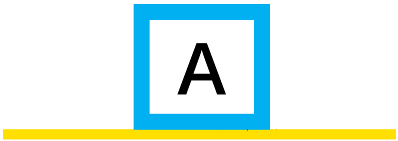
研 究 方 向

- 已知药物，分析靶蛋白（药物 → 靶蛋白）
  - 平台1：靶标捕获
  - 平台2：互作分析
- 已知靶蛋白，分析药物（靶蛋白 → 药物）
  - 平台3：高通量对接筛选
  - 平台4：高通量实体库筛选
- 已知药物及靶蛋白，分析亚结构药效机制（靶 药 → 机制）
  - 平台5：表位捕获平台
  - 平台6：药效分子动力学运算平台
  - 平台7：药物口袋反向验证平台

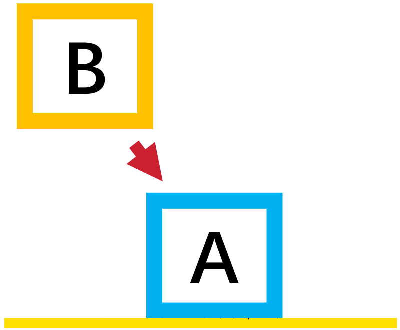
技术交流微信

技术背景：反应平衡

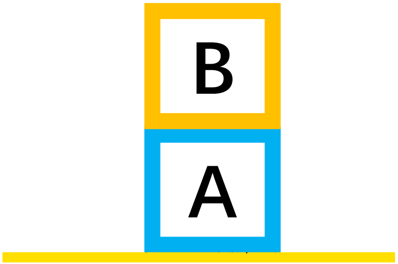




1. 固定A分子



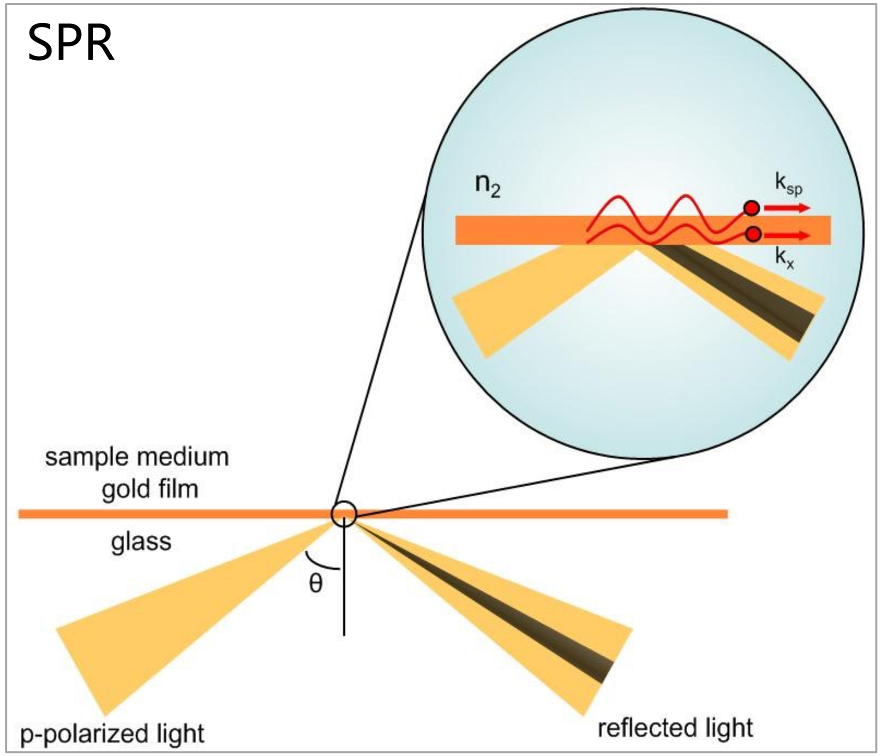
2. B分子结合

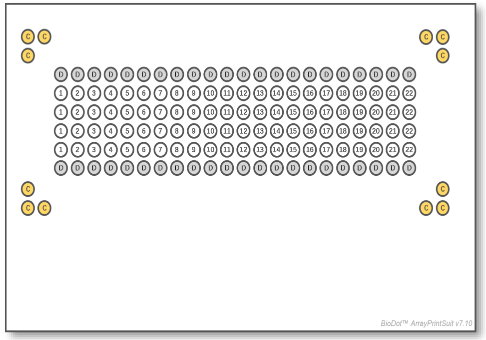


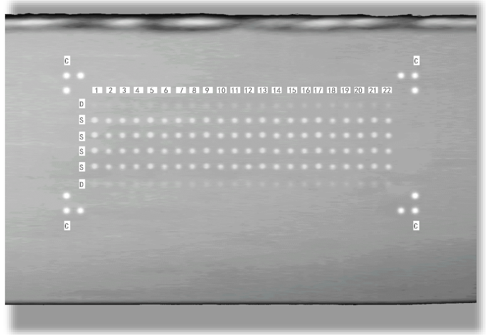
3. 得到AB复合体

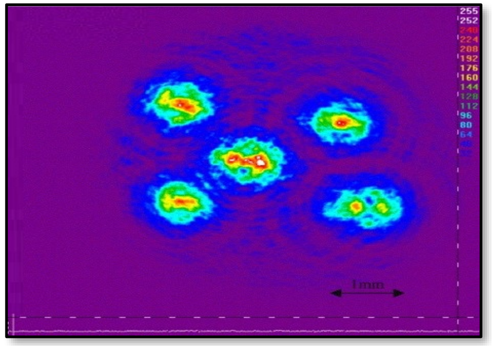
技术基础：表面等离子共振

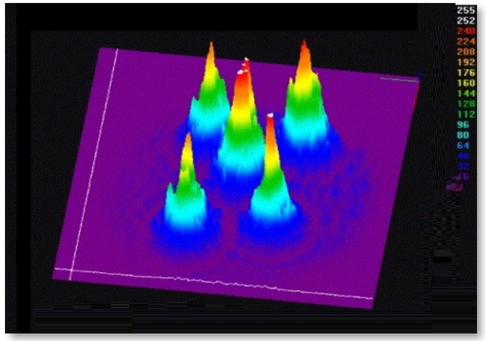
(Surface Plasmon Resonance, SPR)

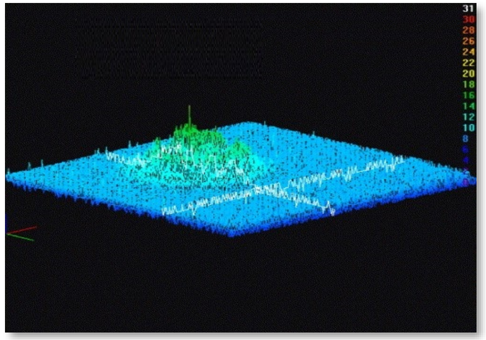


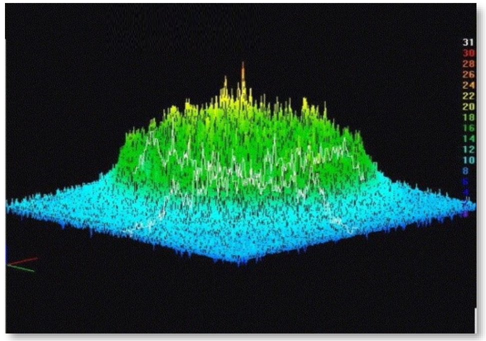

1.阵列打印图纸


2.阵列表面扫描图像


3.样品共振扫描平面图


4. 样品共振强度峰图

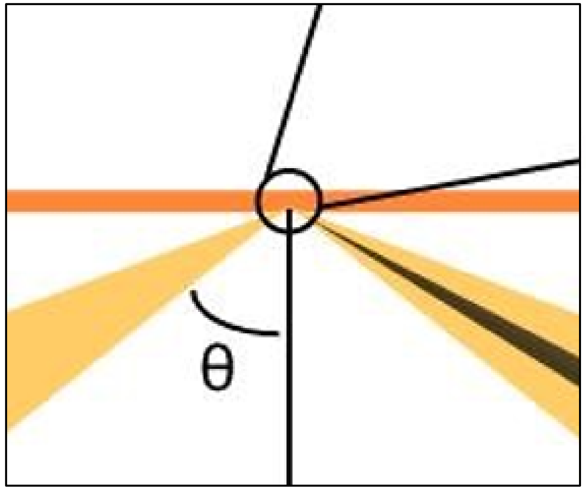

5.基底信号


6. 配体结合信号

技术基础： SPR 技 术 迭 代

第一代 SPR

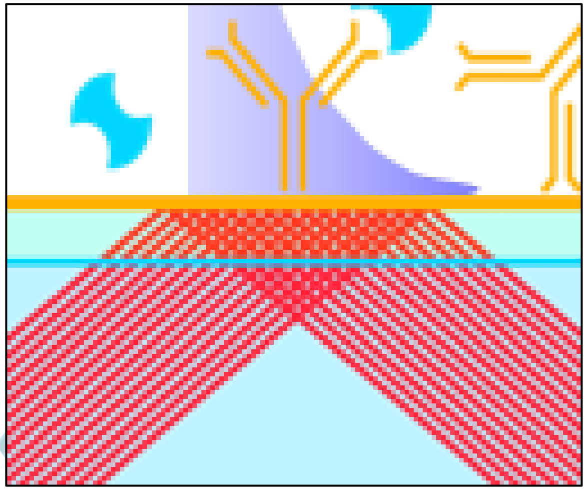
锥形光路



代表设备：
- GE Biacore T200/1K
- 赛多利斯 Octet SF3

第二代 SPR (SPR image，SPRi)

平行光路



代表设备：
- Berthold bScreen LB 991
- Plexera PlexArray H1000